DIET PATH – 2

INDIA TECH CAREER INTELLIGENCE | Hackathon Challenge Report & Execution Guide — EDA Edition

Challenge Attribute	Specification
Challenge Type	Exploratory Data Analysis (EDA), workforce intelligence and evidence-based reporting
Eligible Batch	DIET Path 2
Entry Format	5 students per team
Dataset	India Tech Career Intelligence 2026 CSV; actual supplied file contains 100,000 rows and 33 columns
Expected Scope	Data quality audit; job/internship mix; skills, companies, location, work mode, salary/stipend, experience, applicant and career-growth indicators
Machine Learning	Not required; challenge stops at descriptive EDA
Primary Outcome	Evidence-based career-market insights and practical guidance for learners/job seekers
Status	Before Hackathon Start

Consolidated Hackathon Guide

1. Executive Summary

India Tech Career Intelligence is a business and career-analysis challenge. Teams act as junior workforce analysts and use NumPy, Pandas, Matplotlib and Seaborn to audit the supplied records, understand job categories and roles, identify requested skills, compare opportunity locations and work modes, and describe salary/stipend patterns. Teams should create a chart-backed report that helps students understand where opportunities appear in this dataset and what skills are associated with role categories.

The challenge has eight problem statements, from data validation to a final career-intelligence report. No machine-learning model is built. Findings must be described as patterns in the supplied dataset—not as a complete or verified census of India's technology labor market.

2. Original Business Situation

Students and early-career professionals need practical insight into technology jobs and internships: which roles and skills appear, which companies and locations are represented, how salary ranges and internship stipends vary, and how work mode and experience requirements differ. A career cell wants a concise evidence-based snapshot to support discussion of skill development and job-search strategy. Teams must answer using only the supplied records and clearly state dataset limitations.

3. Official Challenge Statement

Perform a complete exploratory data analysis on the supplied India Tech Career Intelligence dataset using NumPy, Pandas, Matplotlib and Seaborn. Audit data quality; analyse role/job-category and internship opportunities; extract skill demand from the supplied skills field; compare companies, locations, work modes and experience requirements; examine salary and stipend fields; and deliver a final report with evidence-backed career insights. Do not build predictive models or claim the dataset represents all current vacancies.

4. Eight Problem Statements to Solve During the Hackathon

Problem Statement 1 — Audit and Clean the Career Dataset

Load the CSV with Pandas and inspect shape, data types, summary statistics, missing values, duplicates, category consistency and malformed values. Preserve the raw file and work on a separate cleaned DataFrame. Parse posted_date and application_deadline using the documented day-month-year pattern; inspect invalid dates and deadline ordering. Convert career_growth_index from percentage text only after documenting the conversion. Document every cleaning decision in a log.

Problem Statement 2 — Understand Job Types, Categories and Roles

Compare job_category, job_title, employment_type, contract_type and experience_level. Identify the most frequent roles and categories, distinguish internship records from full-time roles using the available fields, and report counts and shares with denominators. Do not assume job title alone fully defines employment type.

Problem Statement 3 — Identify Skills Requested by Employers

Use skills_extracted as the source for skill strings. Split comma-separated skills, normalize whitespace/case carefully, count skill mentions overall and by job_category/job_title/experience_required, and compare skill_count with the parsed list length. Identify common combinations and category-specific skills. Do not use NLP/modeling or claim a skill is universally required; it is only frequent in this dataset.

Problem Statement 4 — Analyse Companies and Industry Representation

Compare company_name, company_type, company_size and industry. Report which companies and company types account for the largest record counts, and compare role/category mix across them. The supplied dataset has one industry value in the observed snapshot, so do not claim broad industry comparison where variation is absent. A company's record count is not proof of its actual hiring volume.

Problem Statement 5 — Map Opportunity Locations and Work Modes

Compare location_city, location_state, location_country and work_mode. Identify locations with the most records and compare remote, hybrid and on-site shares across roles/categories and experience requirements. Report n alongside percentages. The supplied records are India-focused; do not infer location availability beyond the dataset.

Problem Statement 6 — Analyse Salary Ranges and Internship Stipends

Inspect salary_minimum, salary_maximum, salary_currency and stipend_amount. Confirm currency and units before interpreting amounts; use INR records only where supported by salary_currency. Check minimum ≤ maximum, zero/negative values and outliers. Report median and quartiles, range distributions, and salary-range midpoint only as a clearly labelled derived measure. Analyse stipend_amount separately from salary. Compare salary ranges by job category, role, location, work mode and experience only with adequate sample sizes. Do not treat stipend as annual salary or infer take-home pay.

Problem Statement 7 — Examine Experience, Education, Applicants and Career Indicators

Compare experience_required, experience_level, education_required, applicant_count, interview_difficulty, probation_period, shift_type, is_active and career_growth_index where meaningful. Report category counts and missingness. Applicant counts may reflect platform snapshots and are not a measure of hiring probability. Treat career_growth_index as a supplied field whose definition is not documented; do not interpret it as a validated forecast or guaranteed growth. Analyse active status descriptively only.

Problem Statement 8 — Build the Final Career Intelligence Report

Combine findings from Problem Statements 1–7 into a concise dashboard/report. Identify three evidence-backed career insights, such as skill clusters, location/work-mode patterns or salary-range differences. Each insight must include the supporting count/sample size, chart or table, caveat and a practical next step for learners or the career cell. Separate observed patterns from hypotheses and avoid recommendations unsupported by the dataset.

5. Dataset Overview and Suggested Data Dictionary

Field / Group	Dataset columns	Role in Analysis
Role taxonomy	job_title, job_category, technical_domain	Role/category counts and comparisons
Company	company_name, company_size, industry, company_type	Company representation and role mix
Geography & mode	location_city, location_state, location_country, work_mode	Opportunity distribution by location and remote/hybrid/on-site
Compensation	salary_minimum, salary_maximum, salary_currency, stipend_amount	Salary ranges and internship stipend analysis; verify units
Candidate requirements	experience_level, experience_required, education_required	Experience and education requirements
Skills	skills_extracted, skill_count	Skill frequency, category-specific demand, consistency checks
Posting / platform	posted_date, application_deadline, source_platform, applicant_count, is_active	Posting timing, deadline checks, platform and applicant snapshot
Other attributes	employment_type, contract_type, job_benefits, interview_difficulty, probation_period, shift_type, career_growth_index	Descriptive comparisons; field definitions may need caveats
Text field	job_description	Optional qualitative context; avoid exposing HTML/template text or treating it as structured truth

6. Data Definition, Handling and Analysis Rules

- Keep an unchanged copy of the raw CSV; perform cleaning on a separate DataFrame.
- Document missing values, duplicates, type conversion, category treatment and date parsing.
- Do not treat identifiers or categorical codes as numeric measures.
- Split skills_extracted consistently; validate skill_count against parsed skills and document mismatches.
- Salary minimum/maximum and stipend are separate measures. Confirm units and currency before any annual/monthly interpretation.
- Show n beside every percentage and state the denominator.
- Parse dates with errors handled explicitly; inspect posted date versus application deadline.
- Every chart needs a title, labelled axes and legend when needed, plus a 2–3 line interpretation.
- Describe observed patterns rather than causal claims. Avoid claims about hiring success, salary fairness, future demand or career growth without evidence.

7. Tools and Techniques Requirement

- NumPy: use arrays/functions such as np.where, np.nan, np.mean, np.median, np.percentile and np.round in at least three places.
- Pandas: read_csv, info, describe, isnull, duplicated, groupby, value_counts, crosstab/pivot_table, string split/explode, datetime parsing, sorting and filtering.
- Matplotlib: at least three customized charts (bar, histogram, line or scatter) with titles, labels and legends as appropriate.
- Seaborn: at least four charts such as countplot, barplot, histplot, boxplot and heatmap where supported by data types.
- Written interpretation (2–3 lines) under every chart or summary table.
- Scope is restricted to NumPy, Pandas, Matplotlib and Seaborn. Machine learning, prediction, NLP and model building are not required or rewarded.

8. Key Business Metrics for the Analysis

Metric	Business Interpretation
Record / opportunity volume	Total records and counts by job title/category, company, location and work mode
Role and employment mix	Count/share by employment_type, contract_type and internship-related categories where explicitly represented
Skill frequency	Count of postings mentioning each skill; include denominator and skill parsing method
Location mix	Record count and share by city/state; work mode mix by location
Salary coverage and range	Valid salary records, currency, min/max distribution, median and quartiles; report valid n
Internship stipend	Valid stipend count and distribution, analysed separately from salary
Experience/education mix	Counts and shares by experience_required, experience_level and education_required
Applicant snapshot	Distribution of applicant_count; descriptive only, not selection probability
Posting freshness/active	Date coverage, active/inactive counts, posted/deadline checks; do not imply live vacancy status
Career growth indicator	Descriptive distribution of career_growth_index; definition is unspecified and must be disclosed

9. Required EDA

- Dataset shape, data types, descriptive statistics and duplicate checks.
- Missing values, blank strings, invalid dates, category-value audit and salary consistency checks.
- Job title/category and employment/contract type distributions.
- Skill frequency overall and at least two role/category comparisons.
- Company and location/work-mode comparisons.
- Salary range and stipend distributions with valid sample counts and currency/units.
- Experience and education requirement comparisons.
- At least three meaningful segment comparisons and one comparison involving internship records if present.
- Clear distinction between observed patterns, hypotheses and causal claims.

10. End-to-End Execution Workflow

1. Understand the career-intelligence objective and dataset fields.
2. Import NumPy, Pandas, Matplotlib and Seaborn; load the CSV.
3. Audit shape, head, info, describe and data types.
4. Check duplicates, blanks, missing values, category values and numeric ranges.
5. Parse posted_date/application_deadline and document invalid values.
6. Create job-category, title, employment and experience summaries.
7. Parse skills_extracted; compare skill_count and analyse skill frequency.
8. Analyse company, location and work-mode distributions.
9. Validate salary and stipend fields; compute descriptive statistics with clear units.
10. Compare experience, education, applicant and career indicator fields.
11. Build charts, pivot tables and a concise dashboard.
12. Write evidence-linked insights, recommendations, assumptions, limitations and next steps.

11. Segment Interpretation

Segment Type	Meaning	Business Question
High-volume role/category	Large record count or share	Which roles/categories dominate the supplied dataset?
Skill cluster	Skills frequently appearing together or within a category	Which skills could learners investigate for a target role?
Location concentration	Large record count in city/state	Where are opportunities represented in these records?
Remote/hybrid/on-site segment	Different work-mode shares across groups	How does work mode vary by role or location?
Higher salary range segment	Higher observed median/range with adequate n	Which categories show higher listed salary ranges?
Internship segment	Records explicitly classified as internships	What stipend and skill patterns appear in internship records?
Small-sample segment	Few records in a group	Could the result be unstable because n is small?
Data-limited segment	Field absent, unclear units or definition	What additional data is needed before comparison?

12. Business Decision Scenario

Assume a career cell has limited time and wants to prepare a practical student briefing. Teams must use EDA to explain which role families and skills are most represented, where opportunities appear, how work mode and experience requirements vary, and what compensation information can safely be compared. Prioritise meaningful record counts and valid salary/stipend sample sizes, not percentages alone.

- Verify whether records are unique postings and whether repeated listings exist; do not assume one row equals one unique employer vacancy.
- Confirm salary units, period and whether values represent annual/monthly compensation before making comparisons.
- Separate stipend from salary and internship from full-time employment.
- Use location/work-mode patterns as dataset representation, not national labor-market rankings.
- Treat applicant_count and is_active as supplied snapshots; do not infer probability of selection or current availability.
- Treat career_growth_index as undefined metadata unless a source definition is supplied.
- Recommendations should be framed as learning or further-investigation priorities, not guaranteed job outcomes.

13. Technology Stack

- Python: NumPy, Pandas, Matplotlib and Seaborn only.
- Jupyter Notebook or VS Code.
- SQL and machine-learning libraries (such as scikit-learn) are not required.
- Git/GitHub recommended for reproducibility.
- AI assistants may help with code/debugging, but students must validate outputs and explain their analysis.

14. Required Submission Package

Deliverable	Requirement
EDA Notebook	Complete workflow from data loading through final insights

Deliverable	Requirement
Source Code	Clean, commented Python code that runs top-to-bottom
Data Cleaning Log	Issues found, decisions applied, dates/units/category handling
Visualization Set	At least 10 charts using Matplotlib and Seaborn, with titles, labels, n and interpretation
Key Insights Summary	Findings for each of the eight problem statements
Career Recommendations	Practical, evidence-linked learning/job-search investigation steps
README	Setup, dependencies, assumptions, data handling and execution steps
Business Solution	Recommended 5–10 evidence-based next steps/checks

15. Recommended Student Report Structure

13. Executive Summary
14. Business Problem
15. Dataset and Field Definitions
16. Data Cleaning and Quality
17. Role, Category and Employment Mix
18. Skill Demand Analysis
19. Company, Location and Work-Mode Analysis
20. Salary and Internship Stipend Analysis
21. Experience, Education and Applicant Indicators
22. Key Insights and Findings
23. Career/Student Recommendations
24. Limitations and Next Steps

16. Judging Rubric

Dimension	Judging Focus
Problem Understanding	Clear career-intelligence objective and accurate field interpretation
Data Cleaning & Quality	Correct missingness, duplicates, date, salary and category handling; documented decisions
NumPy & Pandas Usage	Correct groupby, crosstab, filtering, parsing, string processing and NumPy functions
Visualization	Chart choice, titles, labels, readability, sample sizes and interpretation
Skill Analysis	Reproducible skill parsing and meaningful role/category comparison
Segment Analysis	Meaningful comparisons across role, company, location, work mode, salary and experience
Insight Quality	Accurate observations, sample sizes and no unsupported causal claims
Career Recommendation	Practical, evidence-linked learning or investigation recommendations
Limitations	Transparent units, field definitions, duplicates, sample size and dataset coverage

Dimension	Judging Focus
Presentation & Q&A	Clear explanation and technical/business defence

17. Challenge Boundaries

- Machine learning and predictive modelling are not required.
- Clustering, deep learning and NLP are not required.
- Use only NumPy, Pandas, Matplotlib and Seaborn for analysis.
- Do not delete or alter the raw data file.
- Do not hide missing values, duplicate listings or data issues; document them.
- Do not call salary or stipend annual/monthly unless the dataset definition supports it.
- Do not treat applicant count as hiring probability or active status as a verified live vacancy.
- Do not present career_growth_index as a validated forecast without its definition.
- Do not present charts without titles, labels, sample sizes and interpretation.
- Do not claim observed patterns are proven causes or that recommendations guarantee employment.

18. Expected Hackathon Outcome

A successful solution will produce a clean, documented career-intelligence analysis. Teams should move from raw listings to validated fields, accurate counts and distributions, meaningful skill and segment visualisations, cautious career-market insights and prioritised student-facing recommendations.

From	To
Raw career records	Cleaned and validated dataset
Job titles and categories	Role and employment mix
Skill strings	Skill frequency and category comparisons
Location and work mode	Opportunity representation by geography/mode
Salary and stipend fields	Valid descriptive compensation comparisons
Charts and tables	Clear career-intelligence insights
Observed patterns	Practical next steps with limitations

19. Dataset-Specific Limitations

- The file is an educational dataset described as aggregated from public sources; provenance, deduplication method and representativeness are not fully documented in the supplied CSV.
- Salary and stipend units/periods are not specified in the field names alone. Do not infer annual/monthly values without source documentation.
- career_growth_index is stored as percentage-like text, but its methodology and meaning are not supplied.
- job_description contains HTML-like markup/template text; do not assume it provides rich, validated job requirements.
- Applicant counts, active flags and source platform are snapshots; they do not prove current vacancy status or hiring outcomes.
- Small segments can show unstable percentages; always report n.
- EDA does not prove why salaries, skills or work modes differ, nor that a particular skill guarantees employment.

20. Final Checklist

- [ ] Raw CSV preserved; cleaned copy created.
- [ ] Actual dataset shape and all field types/missingness/duplicates audited.

- [ ] Salary currency and units limitations disclosed.
- [ ] Skill parsing documented and skill_count checked.
- [ ] All eight problem statements addressed or limitation explicitly documented.
- [ ] At least 10 charts/tables have titles, labels, sample sizes and interpretation.
- [ ] Salary and stipend analysed separately.
- [ ] No unsupported national-market, causation or employment-success claims.
- [ ] Recommendations link directly to evidence and are framed as next steps.
- [ ] Notebook and README run from beginning to end.